Executive Summary

Online consumer-lending fraud has become increasingly subtle and adaptive, with fraudsters deliberately emulating legitimate digital and behavioral patterns to evade detection. Traditional rules and static vendor indicators remain important, but they no longer fully capture the evolving signature of modern fraud. In addition, conducting manual reviews of loan applications incurs significant costs, further constraining operational capacity. Compounding these challenges, Enova can operationally review only about 5% of applicants, creating a stringent requirement that any improvements in fraud detection must not only enhance predictive accuracy but also concentrate risk within a very small, high-quality segment. These dynamics highlight the need for more flexible, behavior-driven features that can reveal identity instability and anomalous activity beyond what existing systems capture.

To address this challenge, our project developed a comprehensive feature-engineering framework focused on uncovering fraud-relevant behavioral signals across multiple internal and external datasets. We engineered over 50 new indicators spanning logical combinations of high-risk conditions, behaviorally grounded rarity and instability measures, and event-based temporal features derived from bank-account and application histories. Each feature was evaluated using missing-value diagnostics, quantile-fraud rate analysis, and a unified performance scoring function to assess its operational usefulness. This systematic evaluation ensured that newly engineered signals introduced meaningful information rather than duplicating the value of Enova's existing vendor and performance attributes.

Integrating the engineered indicators into a CatBoost model produced a significant improvement in predictive performance. Test-set recall increased from 7.96% to 26.92% while accuracy remained near 98%, demonstrating that the new behavioral and temporal features meaningfully strengthened the model without increasing false positives. Feature-importance rankings and decision-path visualizations further revealed how these engineered indicators complement baseline attributes: vendor and performance variables establish broad risk strata, while the engineered features refine risk within those strata by capturing nuanced behavioral patterns such as rapid information changes, unusual timing gaps, or cross-feature inconsistencies. This layered decision structure provides transparency and aligns with the operational expectations of Enova's fraud-modeling teams.

Overall, our findings highlight that intuitive, behaviorally grounded indicators, many derived from simple transformations of existing internal data, can substantially enhance fraud detection in an online lending environment. These features enable earlier recognition of emerging fraud tactics, increase the concentration of fraudulent cases within Enova's 5% manual-review capacity, and provide a foundation for more adaptive and interpretable risk models. The framework developed in this project offers actionable insights for strengthening Enova's fraud-screening infrastructure and serves as a roadmap for future expansions in data diversity, model robustness, and operational integration.

Fraud Prevention in FinTech: Improving Detection Through Innovative Modeling and Feature Engineering

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1 Introduction

Enova International is a leading financial technology company providing fast and accessible credit solutions to consumers underserved by traditional institutions. Operating across multiple proprietary and partner brands, Enova processes a large volume of daily loan applications originating either through paid Lead channels or directly through Organic traffic.

As a digital lender, Enova relies heavily on data-driven systems for underwriting, credit-risk evaluation, fraud detection, and regulatory compliance. Its decisioning infrastructure integrates third-party credit bureaus, identity-verification providers, and internal behavioral data to form a comprehensive view of each applicant.

Since fraudulent applications directly affect profitability and customer trust, accurate and early identification of high-risk cases is essential. Enova continually invests in machine-learning models, workflow automation, and layered human review to mitigate losses from fraud and improve customer experience.

2 Problem Statement

Online consumer-lending fraud is becoming increasingly subtle. Traditional rules and static vendor indicators are less effective against these evolving tactics, leaving only weak signals of identity instability or device irregularities. Enova faces this challenge acutely. Existing rule sets and historical fraud patterns do not generalize well to new behaviors, while manual review is resource-intensive and limited to roughly 5% of applicants. Any improvement to the fraud-detection framework must therefore not only enhance accuracy, but also concentrate risk within a small, high-quality segment that aligns with operational capacity.

This project addresses that need by engineering new, behavior-driven and temporally grounded fraud indicators. These features are designed to reveal subtle anomalies in identity usage and cross-dataset interactions that complement existing models. To evaluate their incremental value, we compare baseline and enhanced machine-learning models under Enova's requirement that features meaningfully separate the highest-risk 5% of applicants.

3 Data

Among all datasets provided by Enova, our analysis primarily relied on three internal sources: the "Complete People" dataset that records basic information of each customer, the "Customer Changes" dataset which records the update history of application-level information, and the "Bank Account History" dataset, which tracks updates to bank account information.

These internal datasets were supplemented with several third-party credit-reporting sources, including "Emailage" for email-based risk signals, "Iovation" for device intelligence, and "RiskView" for behavioral credit data. Merging the internal and external datasets provide a comprehensive view of each applicant, combining static identity attributes with dynamic behavioral indicators that support more effective fraud detection. Finally, a labeled training dataset which includes a fraud indicator was used to evaluate how our engineered features corresponded to known fraudulent behavior.

4 Methodology

The methodological framework consisted of three integrated phases: construction of a baseline model using Enova-recommended features, systematic exploration and engineering of new features across multiple datasets, and evaluation of an advanced model incorporating both baseline and engineered features.

4.1 Baseline Model

The baseline model was built from variables that Enova had already identified as strong historical fraud predictors. These include core application attributes (e.g., loan amount, re-lending status, income frequency), third-party risk scores (RiskView, Clarity Behavior Score, Clear Fraud Insight), static Emailage indicators (score, risk band, domain checks), device and IP attributes (Iovation), as well as banking and geographic summary information (INEXP bad-rate measures, prior loan outcomes, approval state). Together, these features reflect Enova's existing fraud-screening logic and serve as a benchmark against which the incremental value of newly engineered features can be assessed.

To model this baseline feature set, we employ CatBoost, a gradient-boosted decision-tree algorithm well suited to Enova's data. CatBoost natively handles mixed feature types and high-cardinality categorical variables without manual encoding, treats missing values as potentially informative (which is critical when missingness can reflect evasive behavior), and incorporates ordered boosting plus regularization to reduce overfitting in imbalanced settings typical of fraud detection. Below is Table 1 summarizing the performance of our baseline model.

Dataset	**Recall**	**Accuracy**
Train	0.1338	0.9827
Validation	0.0962	0.9774
Test	0.0796	0.9770

Table 1: Baseline Model Performance Summary

As shown in Table 1, the baseline model exhibits stable performance across training, validation, and test sets, with accuracy near 0.98 and recall in the range of roughly 0.08 - 0.13. The high accuracy is largely driven by the imbalanced dataset, as the majority applications were non-fraud cases, whereas the relatively low recall indicates that the baseline recovers only a modest fraction of fraudulent applicants. From a fraud-prevention perspective, this means that relying solely on existing raw predictors is insufficient to concentrate fraud risk into the narrow top five percent segment that Enova can manually review. This gap motivates the construction of additional, more behaviorally expressive features that can better separate the riskiest applicants from the broader population.

4.2 Exploratory Data Analysis and Feature Evaluation

To structure this analysis, we summarize our exploratory and evaluation procedures into three core components that guided feature construction and assessment throughout the project.

- **Missing-Value Analysis**
  Because missing information can itself be a meaningful fraud indicator, Enova emphasizes that NA values should never be treated as neutral or ignored. For each engineered feature, we

compared fraud rates between missing and non-missing observations to determine whether data sparsity reflected underlying behavioral risk rather than random absence. As illustrated in Figure 1, certain features exhibited substantially higher fraud rates in the missing-value group, supporting the notion that missingness may signal abnormal identity patterns or incomplete digital footprints often associated with fraud.

- **Quantile-Based Binning Assessment**
  To evaluate how fraud risk varies across the distribution of each feature, we applied a standardized quantile-binning procedure. By computing fraud rates across five evenly spaced bins, we could assess monotonicity, detect nonlinear or threshold effects, and identify value ranges that disproportionately concentrate fraud cases. This approach, also shown in Figure 1, enabled rapid screening of large numbers of engineered variables and provided a consistent basis for determining whether a feature exhibited operationally meaningful risk separation.

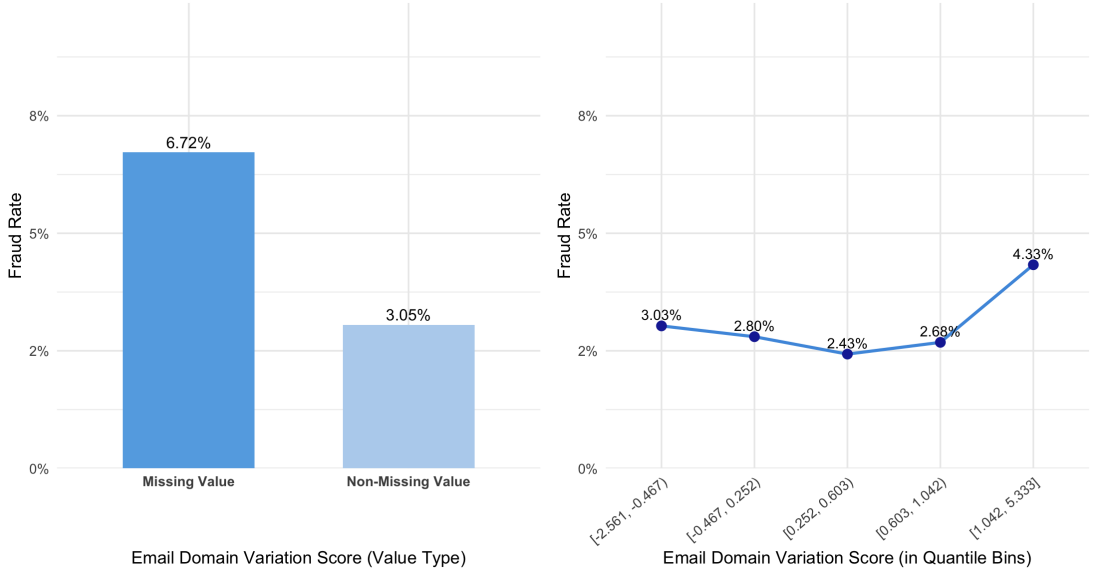

Figure 1: EDA illustration: missing-value effects and quantile-bin fraud rates for the Email Domain Variation Score

- **Evaluation Metrics**
  Given the severe imbalance between fraud and non-fraud cases, accuracy alone is not informative. It is possible that non-fraud applications overwhelmingly dominate the sample, causing even non-predictive features to achieve accuracy above 99%. To address this, we developed a unified evaluation function that reports recall, precision, accuracy, and F1-score for each feature. Particular emphasis is placed on recall, which measures the proportion of fraudulent applications captured within the minority class. This metric aligns with Enova's operational priorities and ensures that feature evaluation remains focused on the identification of high-risk applicants.

4.3 Feature Engineering

Feature engineering formed the core of the project. Our feature-engineering strategy aimed to maintain strong recall to ensure high coverage of known fraud cases while constraining the proportion

of applicants to about 5% of the total applications. This approach ensures that Enova has the manual resources to review the features truly capturing meaningful fraud-related behavior.

To preserve the integrity of our feature-engineering evaluation, we ensured that all engineered features were conceptually and statistically distinct from the baseline feature set. Rather than repackaging information already captured by the baseline, the engineered features were designed to introduce new behavioral signals, particularly patterns of instability, temporal change, or cross-dataset combination that the baseline variables could not represent. Maintaining this separation also helped minimize multicollinearity and allowed us to attribute any performance gains directly to novel information rather than to redundant transformations of existing predictors.

Below, we introduce four of our feature-engineering methods applied in this project and highlight examples that illustrate how these techniques reveal latent behavioral signals in the data.

(1) Cross-Feature Logical Patterns. This method constructs features that activate only when multiple elevated-risk conditions co-occur. Formally, let $\{x_i\}$ be a set of continuous or ordinal features and $\{\theta_i\}$ be a set of *predetermined constant thresholds*. We define binary indicators as:

$$A_i = \mathbb{1}\{x_i > \theta_i\}.$$

From these, we generate composite signals of the form

$$F_{\text{logic}} = A_i \wedge A_j \quad \text{or} \quad A_i \vee A_j,$$

which capture higher-order fraud behaviors not visible from individual variables.

For example, Figure 2 illustrates recall patterns obtained by combining (i) high Emailage/IP risk scores and (ii) frequent email changes over a window size w. We define the specific fixed constants θ_{risk} (risk threshold) and K_{freq} (frequency limit) to form the indicators:

$$A_{\text{risk}} = \mathbb{1}\{\text{risk score} > \theta_{\text{risk}}\}, \qquad A_{\Delta\text{email}(w)} = \mathbb{1}\{\Delta\text{email}(w) > K_{\text{freq}}\}.$$

The joint feature $A_{\text{risk}} \wedge A_{\Delta\text{email}(w)}$ yields substantially higher recall when risk is Emailage-based, reflecting that both Emailage risk and email-change frequency measure instability of the same underlying identity dimension.

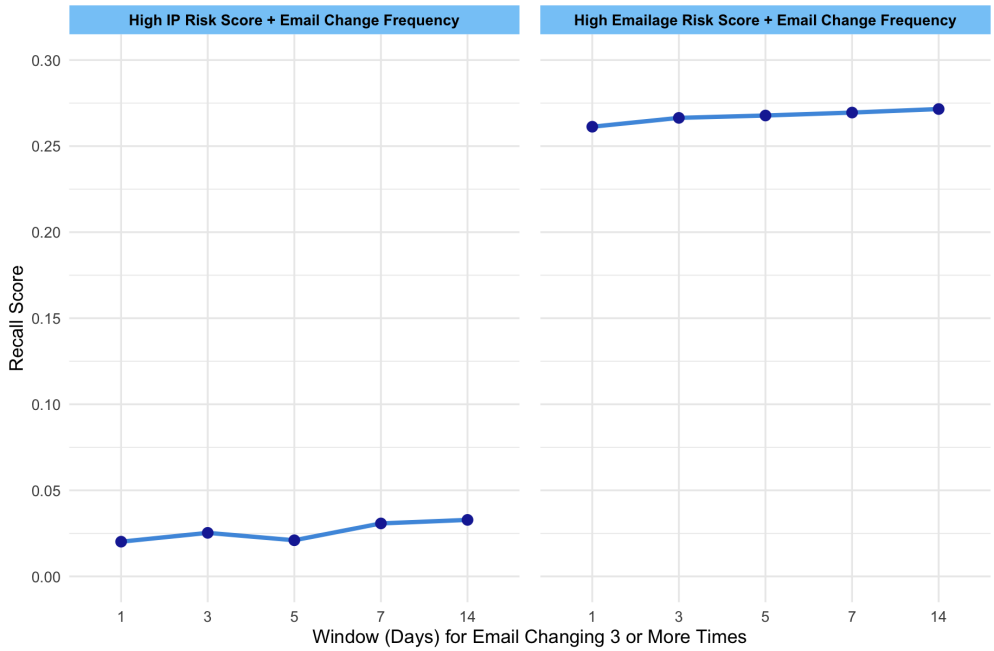

Figure 2: Recall patterns for Emailage/IP risk scores with email-change frequency

In contrast, combining IP-based scores with change frequency provides weaker lift, indicating that not all logical compositions create meaningful fraud separation; the most informative ones align with coherent behavioral mechanisms.

(2) Behaviorally Grounded Indicators. These features model realistic fraud behaviors rather than static attributes. Let each IP-city c have a historical user count

$$U(c) = |\{u : \text{user } u \text{ has appeared from city } c\}|,$$

and define an IP-population rarity flag

$$F_{\text{rare}} = \mathbb{1}\{U(c) < \gamma\},$$

where γ is a rarity threshold. The intuition is that legitimate traffic is distributed across many IP regions, whereas fraud traffic often funnels through VPN/proxy endpoints with low user diversity.

This single feature provides modest but meaningful lift (precision = 0.0287, recall = 0.1266, F1 = 0.0467), consistent with a domain-driven signal that becomes more valuable when combined with other behavioral indicators.

(3) Temporal and Event-Based Features. To extract signal from banking-update behavior, we convert raw timestamps into event-based temporal features. Let t_{app} denote application time and t_{upd} denote a bank-account update. We compute

$$\Delta t = t_{\text{upd}} - t_{\text{app}},$$

and define targeted temporal windows:

$$F_{\text{post}}(w) = \mathbb{1}\{0 < \Delta t \leq w\}.$$

This transformation reveals suspicious patterns, such as bank-account changes shortly after submission, that are not detectable from timestamps alone. By structuring updates into interpretable windows (e.g., $w = 90$ days), the feature captures sequencing behavior indicative of post-application manipulation.

4.4 Advanced Model

After evaluating performance using the selected baseline features, we developed an advanced model that integrates the baseline variables with more than 50 engineered features. Consistent with the baseline framework, the advanced model was trained using CatBoost, leveraging the same algorithmic structure and hyperparameters to maintain strict comparability. By applying the model to the same test set, we ensured a fair comparison between the baseline and advanced models, which enabled us to isolate the predictive gains attributable solely to the newly engineered features.

5 Results

To evaluate the contribution of the newly engineered features, we retrained the CatBoost model using both the baseline variables and the added logical, behavioral, and temporal indicators. Table 2 reports the resulting performance.

Dataset	**Model**	**Recall**	**Accuracy**
Train	Baseline	0.1338	0.9827
	Advanced	0.3121	0.9827
Validation	Baseline	0.0962	0.9774
	Advanced	0.2885	0.9822
Test	**Baseline**	**0.0796**	**0.9770**
	Advanced	**0.2692**	**0.9818**

Table 2: Advanced model performance and comparison with baseline

Compared with the baseline model in Table 1, recall increases substantially across all splits. Accuracy remains essentially unchanged (around 0.98), indicating that the recall gain is not achieved by sacrificing overall predictive stability. Instead, the engineered features improve the model's ability to isolate high-risk applicants while keeping false-positive rates under control.

From a fraud-operations perspective, the improvement in test recall, from 7.96% to 26.92%, shows that the additional indicators introduce new, non-redundant information. The enhanced model is able to concentrate a much larger share of fraudulent applicants within the narrow highest-risk segment required by Enova's 5% manual-review constraint. These results confirm that the new features provide meaningful behavioral and temporal separation beyond what is achievable with the baseline feature set alone.

Among the 73 total features included in the enhanced model, Figure 3 displays the top 40 ranked by CatBoost feature importance. Baseline features are shown in light blue, and the newly engineered indicators appear in dark blue. The plot reveals a clear hierarchy within the model's decision structure: several established vendor and performance variables, such as INEXP bad-rate measures,

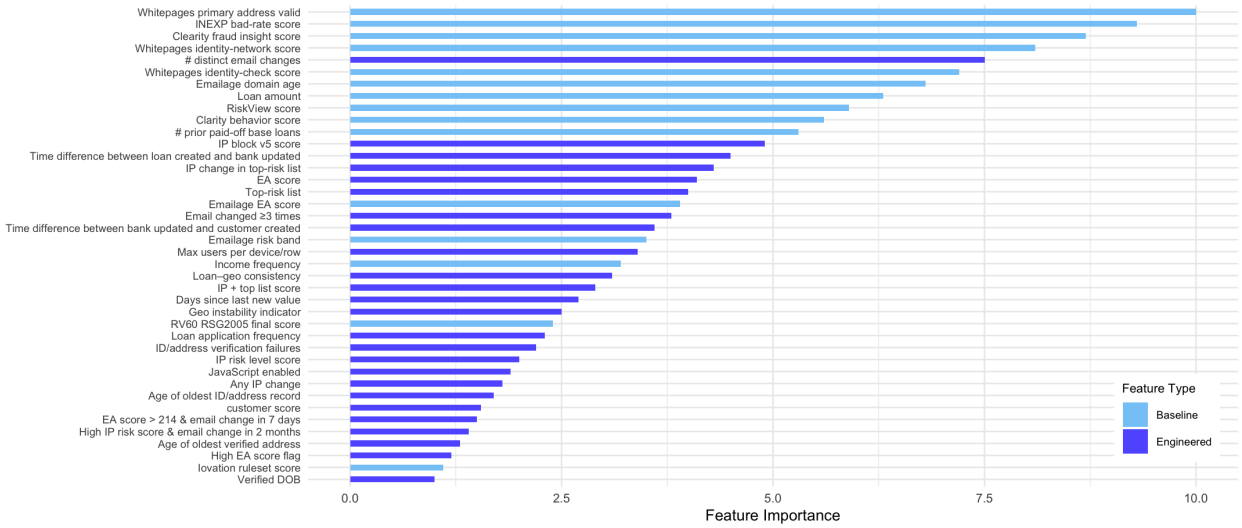

Figure 3: Top 40 feature importances for the advanced model

Whitepages identity and address signals, and Clear Fraud Insight, occupy the very top positions. This confirms that Enova's existing risk infrastructure continues to provide strong predictive value and forms the backbone of the fraud model.

However, the engineered features are not merely marginal add-ons. A substantial number of them appear throughout the top 40, including behavioral indicators (e.g., IP or email change frequency), temporal event-based features (e.g., time differences between bank updates and loan creation), and composite logical patterns. Their consistent presence among the mid- to high-ranking features shows that they introduce meaningful information rather than duplicating the baseline logic.

To illustrate how this blended structure materializes in individual trees, Figure 4 shows two representative decision paths extracted from the CatBoost ensemble. Baseline vendor and performance attributes define broad risk strata, while engineered behavioral and temporal indicators provide fine-grained adjustments within those strata.

In Path A, established vendor and performance variables, *Loan Amount*, *Number of Previously Paid-Off Loans*, *RiskView Score*, and *Clear Fraud Insight Score*, first identify a segment of applicants with seemingly favorable traditional risk indicators. Only after defining this broad, low-risk group do the engineered signals, such as the *IP-Score Prediction* and the *Time Between Bank Update and Customer Creation*, adjust the leaf score, capturing more subtle patterns related to IP-behavior and the sequencing of bank-account changes.

Representative decision paths from the CatBoost ensemble

Path A (leaf score: -0.0191)

- IF *Loan Amount* > 525
- AND *IP Risk Model Score* ≤ 2.5
- AND *Count of Previously Paid-Off Loans* ≤ 18
- AND *RiskView Score* > 648.5
- AND *Time Between Bank Update and Customer Creation* ≤ 5484.5
- AND *Clear Fraud Insight Score* ≤ 739.5

Path B (leaf score: -0.0302)

- IF *SSN Verified Flag* $= 0$
- AND *Time Between Loan Creation and Bank Update* $\leq$ -624.5

Figure 4: Interactions between baseline vendor attributes and engineered indicators

Similarly, Path B combines a baseline rules-engine flag, *Verified SSN*, with an engineered temporal feature, *Time Between Loan Creation and Bank Update*. An unverified SSN becomes more concerning when paired with an unusually short delay between loan creation and the applicant's bank update; the joint condition produces a substantially negative leaf score.

These paths concretely illustrate the interaction pattern suggested by the global importance plot: the enhanced model does not replace Enova's risk infrastructure, but uses engineered behavioral and temporal indicators to refine decisions within the regions defined by strong baseline vendor signals. This interaction mechanism is consistent with the observed performance gains in recall without loss of accuracy, and with the operational goal of concentrating fraud within the top five percent review segment.

6 Discussion

Business Motivation for Prioritizing Recall

In fraud detection, the costs associated with misclassification are fundamentally asymmetric. A false negative, failing to flag a truly fraudulent applicant, directly exposes the lender to financial losses and increased downstream collections activity. In contrast, a false positive, incorrectly flagging a legitimate applicant, typically results only in additional manual review or minor customer inconvenience. Because these soft costs are far less severe and more controllable operationally, prioritizing the identification of fraudulent cases is essential. For this reason, recall, defined as the share of true fraud cases successfully captured by the model, is the most business-critical metric for evaluating feature effectiveness in our project.

Model Robustness and Real-World Interpretation

We deliberately used a conservative re-weighting strategy to avoid overfitting to short-lived fraud patterns observed in the historical data. Enova noted that several behaviors in the sample, such as repeated use of specific IP addresses, had already shifted in production, so aggressively up-weighting

fraud cases could have made the model highly sensitive to distribution drift. Instead, we promoted stability through lighter fraud weights, downsampling of the non-fraud majority class to improve balance, and early stopping once validation performance stopped improving.

Even with these safeguards, the reported gains should be interpreted with caution. The dataset contains relatively clean and homogeneous fraud signatures, allowing the model to lean on a small set of dominant features, and some of these signals have weakened or changed in production. As a result, our metrics likely represent an optimistic upper bound on real-world performance. This does not diminish the value of the engineered features: under the dataset's conditions they materially enhance predictive power and help isolate high-risk applicants within the targeted 5% review group. Rather, it underscores the need for ongoing data diversification, periodic retraining, and regularization to maintain robustness as fraud tactics evolve.

7 Conclusion

This project shows that engineered behavioral, temporal, and cross-feature indicators can enhance Enova's fraud-detection capabilities beyond what established vendor variables provide. The baseline model captures historical fraud patterns but struggles to isolate risk within the 5% manual-review segment due to increasingly subtle behaviors. By modeling identity instability, rapid information changes, and cross-dataset inconsistencies, the enhanced model increases recall from 7.96% to 26.92%, demonstrating that these features add genuinely new signal. Decision-path analysis further illustrates how the model refines risk within strata defined by strong vendor attributes, creating a layered and interpretable decision structure.

Overall, the results highlight the operational value of flexible, behavior-driven features. They provide a foundation for more adaptive fraud models, earlier detection of anomalous identity patterns, and more efficient use of Enova's limited manual-review capacity.

8 Future Work

Stronger Regularization and Model Interpretability

A valuable direction for future work is to improve model robustness and interpretability through stronger regularization and deeper analysis of tree structures. Techniques such as tighter depth limits, feature penalties, or monotonicity constraints would help stabilize performance as fraud behaviors evolve. In parallel, examining how the fitted trees combine behavioral signals can reveal consistent fraud patterns and how they shift over time. Monitoring these structural changes across retraining cycles could provide an early-warning system for emerging fraud tactics, enabling proactive adjustments to Enova's detection strategy.

Integrating Optimization for Business Value Estimation

Another direction is to build an optimization layer on top of the predictive model to quantify potential business gains. This framework could incorporate an estimated cost matrix for true positives, false positives, true negatives, and false negatives, allowing the system to formally optimize fraud-flagging decisions based on their expected financial impact. Integrating this optimization layer would create a clear bridge between predictive performance and economic outcomes, strengthening the model's relevance to real-world decision processes.